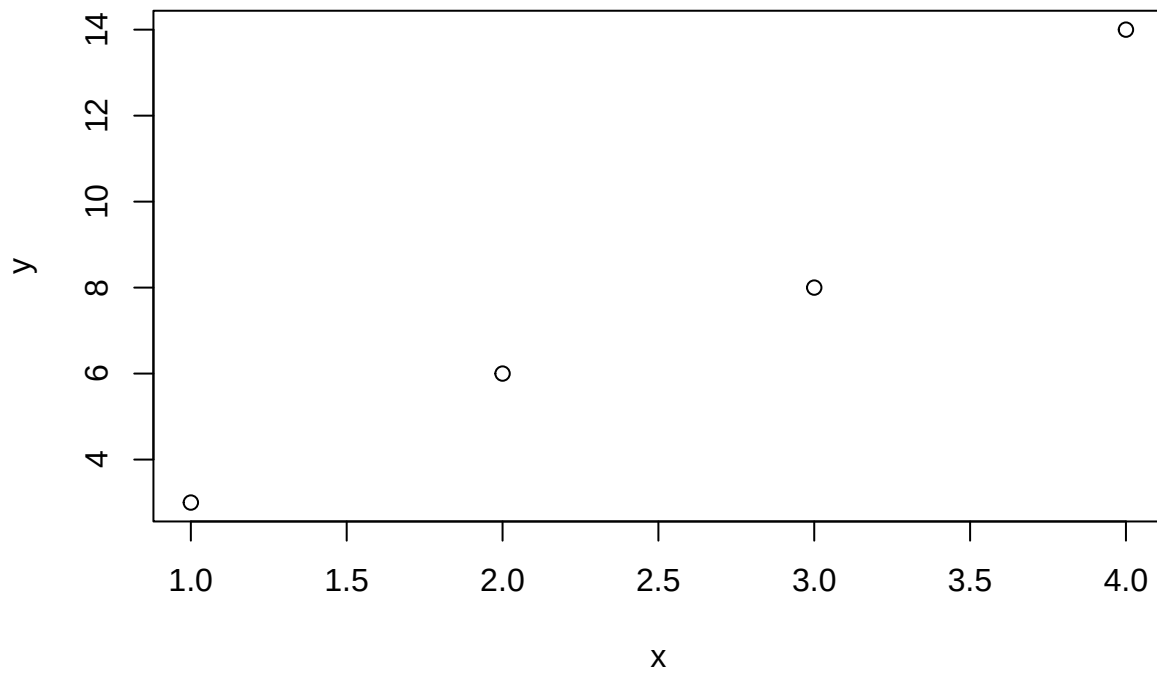


Linear regression

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```
x <- 1:4
y <- c(3, 6, 8, 14)
X <- matrix(c(x^0, x^1), nrow=4)
plot(x, y)
```



In mathematical notation

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, y = \begin{bmatrix} 3 \\ 6 \\ 8 \\ 14 \end{bmatrix}$$

For a linear regression, the design matrix, $\mathbf{X}$ is:

$$\mathbf{X} = \begin{bmatrix} x_1^0 & x_1^1 \\ x_2^0 & x_2^1 \\ x_3^0 & x_3^1 \\ x_4^0 & x_4^1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

The best estimate of the coefficients is:

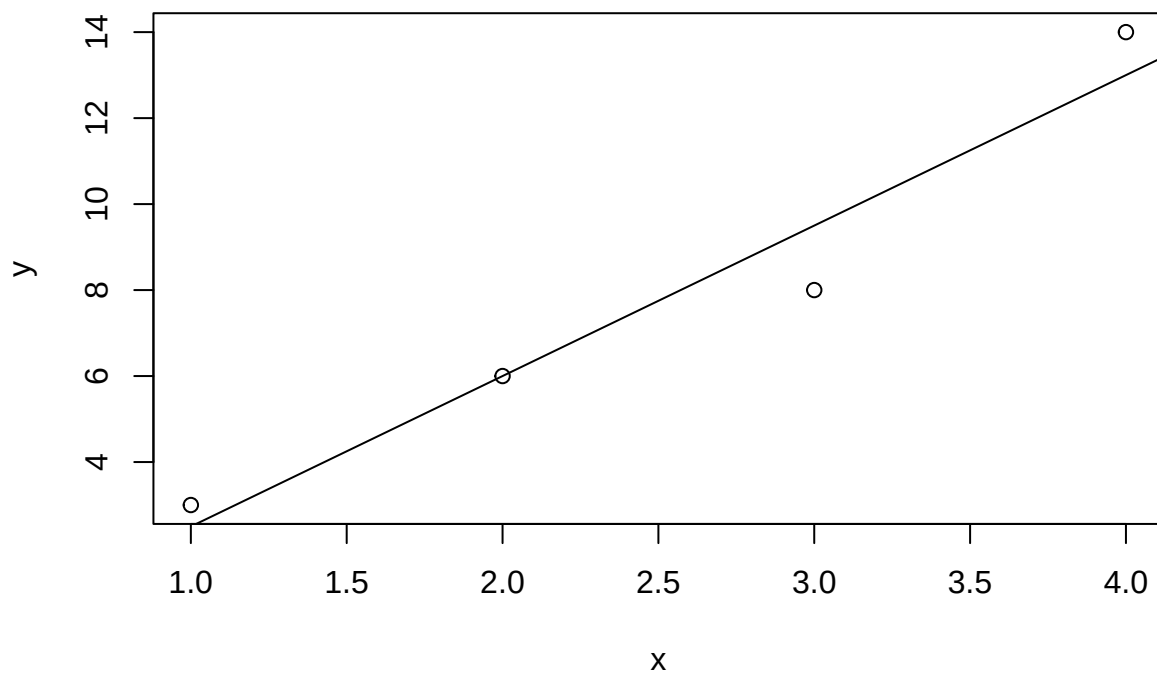
$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y$$

In steps:

$$\mathbf{X}^T y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 8 \\ 14 \end{bmatrix} = \begin{bmatrix} 31 \\ 95 \end{bmatrix} \quad \mathbf{X}^T \mathbf{X} = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix} \quad (\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{\det(\mathbf{X}^T \mathbf{X})} \text{adj}(\mathbf{X}^T \mathbf{X}) \det(\mathbf{X}^T \mathbf{X}) = (\mathbf{X}^T \mathbf{X})_{1,1}(\mathbf{X}^T \mathbf{X})_{2,2} - (\mathbf{X}^T \mathbf{X})_{1,2}(\mathbf{X}^T \mathbf{X})_{2,1}$$

Our fit thus becomes

```
plot(x, y)
abline(a=-1, b=3.5)
```



which we also get by:

```
print(lm(y ~ x))

##
## Call:
## lm(formula = y ~ x)
##
## Coefficients:
## (Intercept)          x
##      -1.0         3.5
```

```
print(t(X) %*% X)
```

```
##      [,1] [,2]  
## [1,]    4   10  
## [2,]   10   30
```

```
print(solve(t(X) %*% X))
```

```
##      [,1] [,2]  
## [1,]  1.5 -0.5  
## [2,] -0.5  0.2
```

```
print(determinant(t(X) %*% X, logarithm = FALSE)$modulus[1])
```

```
## [1] 20
```

```
print(solve(t(X) %*% X) %*% t(X) %*% y)
```

```
##      [,1]  
## [1,] -1.0  
## [2,]  3.5
```